

Cross-pipeline test log

Group code AI-G03. Owner: Member 4, Clustering and QA engineer, Mohammed Osama Hasan (25/27014). Checked on 1 October 2026.

This log records the checks that can be run now. The tests build temporary tables. They do not use, replace, or stand in for the lecturer CSV.

Command and result

```
python -m unittest discover -s tests -v
```

Result: **26 tests, OK**, in 1.244 seconds. Exit code 0.

Module	Tests	Result	What was protected
Data pipeline	6	OK	Schema, fingerprint, missing values, feature matrix
Regression	7	OK	NumPy gradient, training-only scaler, seed, loss plot
Classification	6	OK	Stratified split, training-only scaler, confusion matrix
Clustering	7	OK	k from 2 to 5, one label per row, no target inputs

Leakage checks

- `record_id` stays an identifier. It is not a column of the feature matrix.
- `actual_yield_kg` is the regression target only. Classification rejects it as a feature. Clustering rejects it as an input.
- `dispatch_attention` is the classification target only. Clustering rejects it as an input. A value other than 0 or 1 fails the data check.
- `src/regression.py` does not import scikit-learn and does not call `LinearRegression`. The weight update is batch gradient descent in NumPy.

Scaler checks

- Regression and classification fit the feature mean and scale on the training rows only. The tests compare that mean with the training slice and with the full table.
- Clustering is unsupervised, so its scaler is fit on every row that is being clustered. Those rows are still input measures only.
- The recorded seed in each metrics builder is 42. Repeating a fit with the same seed repeats the regression weights, the classifier coefficients, and the cluster labels.

Each check by name

Data pipeline

- test_feature_matrix_excludes_identifier_and_targets
- test_invalid_dispatch_label_is_rejected
- test_missing_and_duplicate_values_are_reported
- test_report_json_contains_descriptive_statistics
- test_sha256_matches_raw_file_bytes
- test_unexpected_column_is_rejected

Regression

- test_analytic_gradient_matches_a_numerical_check
- test_bad_learning_rate_is_rejected
- test_loss_falls_and_test_metrics_are_finite
- test_module_does_not_call_a_library_regressor
- test_outputs_include_seed_metrics_and_loss_plot
- test_same_seed_repeats_the_weights
- test_scaler_uses_training_rows_only

Clustering rule checked here

Scores are computed for $k = 2, 3, 4$, and 5 . The selected k is the highest silhouette. If two scores tie, the smaller k is kept. `test_tie_keeps_the_smaller_k` locks that rule. Every input row in the synthetic table receives one label, and `clusters.csv` has one line per record id. The metrics text states that a cluster is not a verified real-world category.

The classification scores use positive class 1. The confusion matrix is ordered with labels 0 then 1. A single-class training table is rejected rather than scored.

Classification

- test_outputs_include_seed_metrics_and_confusion_matrix
- test_same_seed_repeats_the_coefficients
- test_scaler_uses_training_rows_only
- test_single_class_training_is_rejected
- test_split_is_stratified_and_target_is_attention
- test_yield_column_is_rejected

Clustering

- test_every_row_is_labeled_from_inputs_only
- test_new_row_matches_the_nearest_saved_center
- test_outputs_include_scores_labels_and_plot
- test_same_seed_repeats_the_labels
- test_scaler_uses_the_clustered_rows
- test_targets_are_rejected
- test_tie_keeps_the_smaller_k

What this run does not prove

The 26 tests passed on synthetic and temporary files. The assessor's command is a different check, and it cannot be run in this repository yet.

Assessor check	Status on 1 October 2026
Clean install from requirements.txt	Blocked. requirements.txt is not in the project yet.
<code>python run_all.py --data data/AI_A1_G03.csv --output artifacts/ --group AI-G03</code>	Blocked. run_all.py and the lecturer CSV are not in the project yet.
Hidden CSV with the same columns	Not run. No issued file is available to copy the schema from.
predict.py with one valid record and one malformed record	Blocked. predict.py is not in the project yet.
Artifacts produced by the command, not written by hand	The unit tests write JSON and PNG only inside a temporary folder, then delete that folder.

QA note for the later full run

- The fingerprint in data_report.json must be the SHA-256 of the issued file bytes, and it must match the value entered on Moodle.
- Row counts, metrics, predictions, and cluster labels must change when the rows change. None of those values may be typed into the source.
- Regression loss and test predictions must be finite. Classification metrics and the confusion matrix must be recomputed. Clustering must show a score for every k from 2 to 5 and a label on every input record.
- A malformed predict.py record must return a clear validation error and no result JSON.

Member 4 evidence. Mohammed Osama Hasan owns this log and src/clustering.py. Export this file as evidence/TEST_LOG.pdf. Run the same unittest command again after run_all.py exists, and add that result to a new entry in this log.